

Technology Architecture / Technical Approach

How ASTRA is built, what it uses, how its AI works, and how it stays secure

Keeps working when parts fail

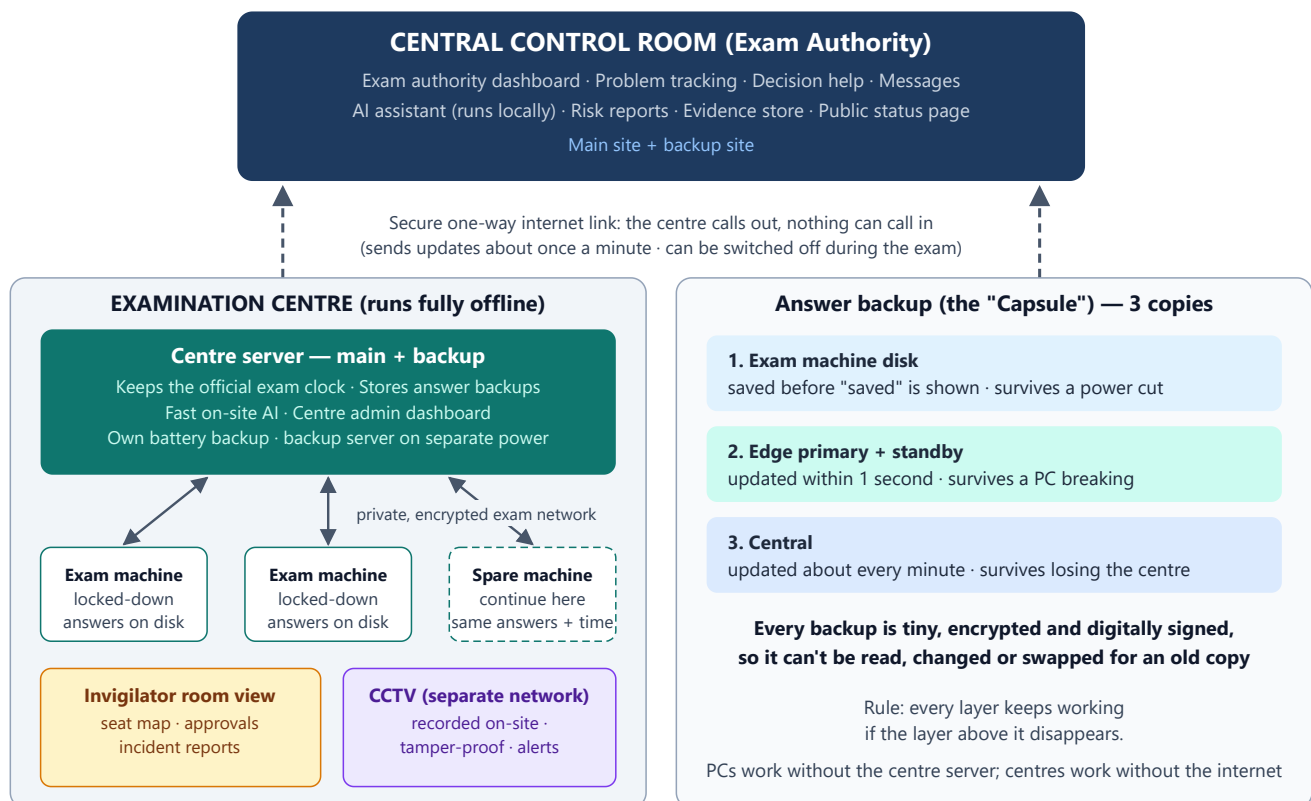
Works offline inside the centre

One tamper-proof record of everything

Trusts no single person or machine

100% free, open-source software

1. How the system is built



Part	What runs there	What it does
Exam PC	Locked-down Linux (only the exam can run) + exam app that works offline	Shows the exam, saves every answer to disk, sends health signals
Centre server (main + spare)	Exam service, answer backups, official clock, action steps, fast AI, dashboards, CCTV recorder	Runs the whole centre on its own; checks every saved answer; sends summaries to the control room
Control room (main + backup site)	Problem tracking, decision help, messaging, assistant AI, reports, evidence store	Oversees all centres, supports decisions, informs candidates, finds repeat problems

Key terms in plain English

Term	Meaning
Centre server (edge server)	A small server inside each exam centre that runs the exam locally. Every centre has two: a main one and a spare
Capsule	A tiny, locked backup of one candidate's exam: answers, question order and time left
Encryption	Scrambling data so it can't be read without the right key
Digital signature	A mathematical seal that proves who created data and shows if anyone changed it
Hash / fingerprint	A short code calculated from data; if the data changes even slightly, the code changes
Tamper-proof record (hash chain)	A log where each entry includes the previous entry's fingerprint, so any edit breaks the chain
Session lock (lease)	A rule that only one PC can use a candidate's exam at a time
Health signals (telemetry)	Automatic status reports from computers: power, network, speed, errors
Failover	Switching automatically to a spare when something fails

2. The answer backup (Capsule)

1–3 KB

for a full exam; each answer change is tiny

AES-256

encryption (standard banking-grade) + digital signature

Linked

numbered and chained, so a missing, edited or old copy is spotted

3 copies

PC disk · centre servers (< 1 s) · control room (~1 min)

- **What it holds:** a candidate code (no names), question order, answers, current question, active exam time, and allowed time.
- **Saving an answer:** saved to the PC's disk → "saved" shown → copied to both centre servers, which check and confirm it within 1 second → summaries sent to the control room about once a minute.
- **Continuing after a failure:** the candidate logs in again and the invigilator approves → the latest valid backup is loaded → the old PC is locked out → **time left = allowed time – time actually used**.
- **Question paper:** delivered to centres in advance, locked. The unlock key is split between several officials in different offices, so no single person can open it early or hold it up.

3. Technology used (all free and open-source)

Area	Technology	What it's for
Exam app	React + TypeScript web app that works offline	The exam screen; runs even without a connection
Saving answers	Browser database (IndexedDB) with forced disk writes	Answers survive sudden power loss
Exam PC setup	Minimal Debian Linux in kiosk mode + firewall	Nothing but the exam can run; PC can only talk to the centre server
Servers	Python (FastAPI) + SQLite / PostgreSQL	Centre and control-room services and databases
Security	Encrypted connections (TLS), digital signatures (Ed25519), one-time codes (TOTP)	Safe communication, proof of origin, secure logins

Area	Technology	What it's for
Dashboards	One web app with different views per role	Invigilator, centre head and exam authority screens
CCTV	Standard IP cameras, MediaMTX, FFmpeg, OpenCV	Live video, fingerprinted recordings, camera-tamper alerts
AI and data	LightGBM, scikit-learn, SHAP, OR-Tools	Predictions, pattern spotting, explaining results, planning seat moves
Fast AI	Small open decision model (e.g. Laya)	Quick, offline yes/no and category decisions
Assistant AI	Small open language model run locally (Ollama)	Writing explanations and messages; no paid online service
Power	Network UPS Tools	Reads battery-backup status

4. How the AI works: two layers

⚡ **Fast AI: inside the centre, works offline, answers in milliseconds**

- **Failure predictor:** chance of a breakdown in the next 5–15 minutes, with the reasons
- **Problem sorter:** what kind of problem it is and how serious
- **Report reader:** turns an invigilator's Hindi or English message into a structured alert
- **Unusual-activity spotter:** e.g. answers entered with no keyboard or mouse use

💬 **Assistant AI: in the control room, not needed to run the exam**

- **Explains** each recommended decision in plain words
- **Estimates** how long a repair will take and **plans** moving candidates to free seats
- **Ranks risky centres** and **finds repeat problems**
- **Drafts** messages and reports in Hindi and English

The exam never waits for AI, and AI never acts alone.

Every AI result shows its reasons · there's a simple rule-based backup for each model · officials approve important decisions · no personal data goes to the AI

Training data: real exam-centre data isn't public, so the AI is first trained on realistic simulated centres (including faults, false alarms and noise). It's then retrained on real data during the trial phases.

5. Dashboards and CCTV

- **Levels:** Exam → Region → Centre → Room → Seat → Candidate. Each person sees only their own level.
- **Grouped alerts:** one vendor failure appears as one problem, not hundreds.
- **Silence is an alarm:** out-of-date data turns grey, and if a centre stops reporting, an alarm goes off.
- **No answers visible:** nobody can see candidates' answers during the exam.
- **CCTV:** cameras use a separate network and record inside the centre. Every 10-second clip is fingerprinted. Alerts fire if a camera is switched off, covered, moved or looping old footage. The control room can view cameras on request.

6. Speed and scale targets

< 10 s

to detect a room-level problem

< 30 s

to send the first message to candidates

< 90 s

for a candidate to continue on another PC

0

effect on the exam if the control room goes down

Each centre runs independently, so adding centres doesn't slow others down. **Under heavy traffic:** saving answers always comes first, less urgent updates slow down automatically, and centres send data in turns instead of all at once.

7. Network safety

Exam PCs have **no internet access**; they can only talk to the centre server. The centre server has a **one-way secure link**: it can send data out, but nothing from outside can connect in. For extra-sensitive exams, the link can be **switched off during the exam** and the data sent afterwards.

8. Security: about 80 attack methods studied

Threat	Protection
Breaking out of the exam screen, installing remote-control software	Locked-down PC setup; PC can only reach the centre server
Someone continuing another candidate's exam	Candidate PIN + invigilator approval; the old PC is locked out
Dishonest centre staff	"Blind courier" server (can't open or change answers); lost-time credits capped; two-person approval
Editing logs or deleting CCTV	Fingerprinted, chained records; candidate receipts; a daily public check code
A fake device or fake server on the network	Encrypted connections that verify each device; digitally signed question paper
Stolen dashboard logins	Two-step login; each person limited to their own area; no all-powerful admin
Tricking or corrupting the AI	AI can't take actions; officials approve; results cross-checked
Fake notices and edited screenshots	Official notices carry a verification code
Malicious software updates	Only signed updates, rolled out to a few centres first